

Edge Healthcare ML: Project References

Past GitHub/Kaggle projects and resources for an agentic edge healthcare IoT device, grouped by the signal each uses.
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1. Curated lists (start here)

Resource	What it covers	Link
awesome-healthcare (kakoni)	Open source healthcare software, libraries, tools and resources	https://github.com/kakoni/awesome-healthcare
awesome-healthcare-ai (medtorch)	Open source healthcare tools, algorithms, datasets and research papers	https://gittrend.io/repo/medtorch/awesome-healthcare-ai
awesome-ehr-deeplearning	Papers on EHR mining, machine learning, and deep learning	https://github.com/hurcy/awesome-ehr-deeplearning
awesome-ppg	PPG libraries, datasets, tutorials and papers covering HR, SpO2 and BP	https://github.com/mintisan/awesome-ppg
Awesome-AI-Agents-for-Healthcare	Papers and projects on agentic AI in healthcare, including image analysis, EHR manipulation, counseling, and patient dialogue	https://github.com/AgenticHealthAI/Awesome-AI-Agents-for-Healthcare

2. Cardiac (ECG)

Project	Details	Link
PTB-XL benchmark	Reproduces the PTB-XL ECG benchmarks and lets you benchmark your own model within the same framework	https://github.com/helme/ecg_ptb-xl_benchmarking
Seq2seq heartbeat classification	Inter- and intra-patient heartbeat classification evaluated on MIT-BIH	https://github.com/SajadMo/ECG-Heartbeat-Classification-seq2seq-model
Edge CNN-Transformer	Hybrid 1D-CNN + Transformer trained on MIT-BIH, quantized to INT8, deployed on Arduino Nano 33 BLE Sense via TFLite Micro	https://github.com/ayekhan12/ecg-arrhythmia-edge-classifier
ESP32 ECG system	ESP32 + AD8232 acquisition, BLE streaming, Android app, and a TFLite arrhythmia model	https://github.com/dimitrios01/ECG-Monitoring-System
GitHub topic: arrhythmia-classification	Includes repos that deploy an MLP and decision trees on ESP32-S3 and Arduino UNO using the Chapman ECG dataset	https://github.com/topics/arrhythmia-classification
ArrythML (paper)	Autoencoder models evaluated as float, INT8 TFLite, and on-device on ESP32-S3	https://arxiv.org/pdf/2606.02256

3. PPG / vitals

Project	Details	Link
E2E-PPG	Pipeline covering filtering, signal quality assessment, reconstruction, peak detection, and HR/HRV computation	https://github.com/HealthSciTech/E2E-PPG
HR from PPG + accelerometer	Udacity AI for Healthcare capstone on wrist PPG pulse-rate estimation	https://github.com/ElliotY-ML/Heart_Rate_Estimation_PPG_Acc
PPG-DaLiA dataset	2.7 GB dataset; scientific, non-commercial use only	https://ubi29.informatik.uni-siegen.de/usi/data_ppgdalia.html
PULSE (paper)	Temporal convolutions plus multi-head cross-attention fusing PPG and IMU data	https://arxiv.org/abs/2210.11415v1

4. Stress / affect

Project	Details	Link
StressAffectDetection	Code for "Stress and Affect Detection on Resource-Constrained Devices" using WESAD, SWELL-KW, DREAMER	https://github.com/nandakrishna/StressAffectDetection
WESAD (Empatica E4)	Uses the wrist device's EDA, BVP, ACC, TEMP plus chest respiration	https://github.com/WJMatthew/WESAD
Stress from ECG	Extracts HRV features with HeartPy from 20 s windows of WESAD ECG	https://github.com/Edouard99/Stress_Detection_ECG

5. Motion (falls, activity, Parkinson's)

Project	Details	Link
TI fall detection	Runs on MSPM0G5187 with NPU using SisFall; reports 97.65% accuracy with under 1 ms inference	https://software-dl.ti.com/C2000/esd/mcu_ai/user_guide/examples/fall_detection_classification.html
Edge Impulse fall detection (Transformer)	SisFall converted and uploaded to Edge Impulse; the project was made public for cloning	https://docs.edgeimpulse.com/projects/expert-network/fall-detection-with-transformer-model-arduino-giga-r1
CNN HAR	Shallow CNN plus statistical features; about 97% on UCI with 128-sample segments	https://github.com/aiff22/har
Parkinson's freezing of gait (Kaggle)	MJFF/Kaggle competition on wearable accelerometer data; 1,379 teams submitted 24,862 solutions	https://github.com/AniMilina/Parkinson-s-Freezing-of-Gait-Prediction
FOG 8th place (paper + code)	Simple 1D-CNN that ranked 8th out of 1379 teams	https://arxiv.org/abs/2307.03475v1

6. Neuro and sleep

Project	Details	Link
TinySleepNet	Efficient sleep stage scoring from raw single-channel EEG	https://www.github.com/akaraspt
TorchEEG	Ready-made TinySleepNet implementation in PyTorch	https://torcheeg.readthedocs.io/en/stable/generated/torcheeg.models.TinySleepNet.html
Seizure prediction (CHB-MIT)	Comparative study of deep learning methods for the pre-ictal period	https://github.com/mnsosa/SeizurePrediction
GitHub topic: seizure-prediction	Includes 1st/2nd/3rd place Melbourne-University AES-MathWorks-NIH challenge solutions	https://www.Github.com/topics/seizure-prediction
EEGformer (paper)	Transformer epilepsy detection for low-channel-count wearable devices (ETH Zurich / Bologna)	https://iris.polito.it/bitstream/11583/2978574/1/Transformer_BioCAS_paper-25agosto.pdf
Sleep apnea from single-lead ECG	PhysioNet Apnea-ECG: 70 patients, split 35 train / 35 test	https://www.medrxiv.org/content/10.1101/2021.04.19.21255733.full.pdf

7. Respiratory / audio

Project	Details	Link
Lung sound classification	Uses patient-aware splits to prevent leakage, compares six models, and does patient-level diagnosis	https://github.com/pedramebd/lung-sound-classification
Sound-Dr	Cough and breathing dataset for pneumonia/COVID-19 with baselines against Coswara and COUGHVID	https://github.com/ReML-AI/Sound-Dr
Kaggle Respiratory Sound Database	ICBHI 2017 data	https://www.kaggle.com/vbookshelf/respiratory-sound-database
Google HeAR	Health acoustics foundation model with HF weights, GitHub, and Colab notebooks	https://developers.google.com/health-ai-developer-foundations/hear/model-card

8. Metabolic (glucose)

Project	Details	Link
GluPredKit	Python framework for blood glucose data handling, training, and evaluation (JOSS 2024)	https://github.com/replicahealth/GluPredKit
GluPred	Attention-based network with transfer learning on OhioT1DM, MIT licensed	https://github.com/r-cui/GluPred
DRL glucose prediction	Deep RL framework for single- or multi-step glucose forecasting	https://github.com/PeterDomanski/BG-Prediction-for-Type-1-Diabetics-using-DRL

9. Imaging (camera-equipped devices)

Project	Details	Link
MedMNIST v2	12 2D and 6 3D datasets at 28x28 resolution; data and code public	https://medmnist.com/
Skin lesion, MobileNet	MobileNet on HAM10000 (7 classes), converted to TensorFlow.js	https://github.com/aryanmisra/Skin-Lesion-Classifer
ResNet50 vs MobileNetV2	Transfer learning comparison on HAM10000, both around 90%	https://github.com/temcavanagh/Skin-Cancer-Detection

10. Multi-disease tabular / symptom triage (usable as agent tools)

Project	Details	Link
Kaggle symptom dataset	132 symptom columns mapped to 42 diseases, with train/test CSVs	https://www.kaggle.com/kaushil268/disease-prediction-using-machine-learning
Disease from symptoms	Naive Bayes, Decision Tree, Random Forest, Gradient Boosting on that dataset plus the Columbia symptom KB	https://github.com/neuraljavi/Disease-Prediction-from-Symptoms
MediCare	Streamlit app predicting diabetes, heart disease, or Parkinson's	https://github.com/aman-chhetri/MediCare
AlDiseasePredictor	Covers diabetes, heart, Parkinson's, liver, breast cancer, malaria, and pneumonia	https://github.com/tushark01/AlDiseasePredictor

11. Agentic layer

Project	Details	Link
AgentClinic	Multimodal agent benchmark in simulated clinical settings, including MIMIC-IV cases and HuggingFace model support	https://github.com/SamuelSchmidgall/AgentClinic
MedAgents-2	Multi-agent medical QA with specialized agents, debate, evidence gathering, and consensus	https://github.com/gersteinlab/MedAgents-2
Awesome-Med-Agent	Paper list with code links	https://github.com/richard-peng-xia/Awesome-Med-Agent
Health-LLM (paper)	Evaluates 12 LLMs on wearable data across mental health, activity, metabolic, and sleep tasks; the fine-tuned HealthAlpaca matched much larger models	https://arxiv.org/abs/2401.06866v2
Google HAI-DEF models	Includes MedGemma 4B, MedSigLIP, and HeAR	https://huggingface.co/collections/google/health-ai-developer-foundations-hai-def
Edge Agents runtime	30 MB offline agent runtime for Pi 5 / Jetson Orin Nano with GPIO/MQTT and local SLMs via llama.cpp	https://github.com/ForestHubAI/edge-agents
llama.cpp on Jetson	Container running Llama 3.2 1B Instruct on-device with quantized GGUF models	https://github.com/Advantech-Containers/LLM-Llama.cpp-on-NVIDIA-Jetson

12. Complete IoT health systems (end-to-end references)

Project	Details	Link
ESP32 health monitor	MAX30102 HR/SpO2 plus DS18B20 temperature, MQTT to Mosquitto, Node-RED dashboard	https://github.com/Andreick/ESP32-health-monitor
Arduino monitoring app	Bluetooth to a Python app with anomaly detection and alerts	https://github.com/Karolkalex/monitoring-app-arduino
ESP32 + AWS IoT Core	MQTT to AWS IoT Core with DynamoDB storage	https://github.com/VishnuramNM/IoT-Based-Patient-Health-Monitoring-System-using-ESP32-and-AWS-IOT-core

13. Toolchains for MCU deployment

Tool	Details	Link
TI tinyml-tensorlab	Classification, regression, forecasting, anomaly detection across 24+ TI MCUs with 31 examples; ships a Claude Code agent skill plugin	https://github.com/TexasInstruments/tinyml-tensorlab
TFLM Arduino examples	Google/Antmicro TFLite Micro demos for Nano 33 BLE Sense, runnable in Colab	https://github.com/antmicro/tensorflow-arduino-examples

Things to watch for

- **Accuracy numbers are often inflated.** Many hobby repos split data by sample, not by patient. The lung-sound repo explicitly flags bias behind reported 95-99% accuracies. Prefer patient-wise splits, such as the MIT-BIH inter-patient setup.
- **Foundation models may not fit on the device.** HeAR's model card lists its current size as too large for on-device deployment. Use it for teacher/embedding work, then distill.
- **Check licenses.** PPG-DaLiA is non-commercial, and MIMIC-based data needs PhysioNet credentialing.

Which sections to start with, by sensor

If the device has...	Start with sections
ECG front-end (AD8232 etc.)	2, 4, 6 (sleep apnea)
PPG (MAX30102)	3, 4, 12
IMU	5
Microphone	7
EEG	6
Camera	9
Pi/Jetson-class host	11 (agent layer on top of any of the above)